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(54) **CRYPTOGRAPHIC SYSTEM, ENCRYPTION
DEVICE, DECRYPTION DEVICE, AND KEY
GENERATION DEVICE**

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U.S.C. 154(b) by 289 days.

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H04L 29/06 (2006.01)
G06F 21/00 (2013.01)

(Continued)

(52) **U.S. Cl.**
CPC **H04L 9/0618** (2013.01); **H04L 9/0861**
(2013.01); **H04L 9/3066** (2013.01)

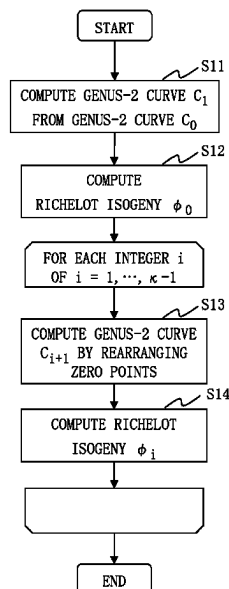
(58) **Field of Classification Search**
CPC ... H04L 9/0618; H04L 9/0861; H04L 9/3066;
H04L 9/0841

See application file for complete search history.

ABSTRACT

A cryptographic system (1) performs a cryptographic pro-
cess in which a Richelot isogeny sequence φ_s whose starting
point is an abelian surface A_0 and whose end point is an
abelian surface A_s is a secret key and the abelian surface A_s
is a public key. An encryption device (28) computes an
abelian surface A_m by transitioning the abelian surface A_s ,
which is the public key, by a Richelot isogeny sequence φ_m
generated by encoding a plaintext m, and sets the abelian
surface A_m as a ciphertext. A decryption device (30) com-
putes a Richelot isogeny φ_m whose starting point is the
abelian surface A_s , which is the public key, and whose end
point is the abelian surface A_m , which is the ciphertext,
based on the Richelot isogeny sequence φ_s , which is the
secret key.

15 Claims, 13 Drawing Sheets



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H04L 9/08 (2006.01)
H04L 9/30 (2006.01)

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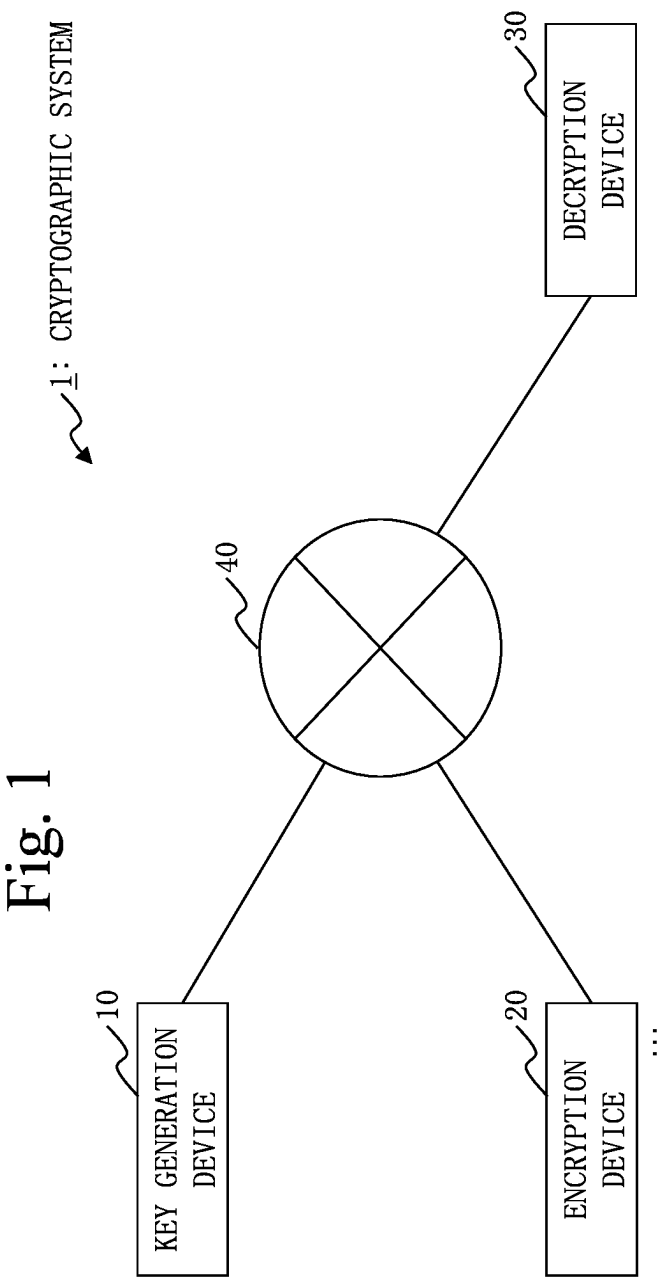


Fig. 2

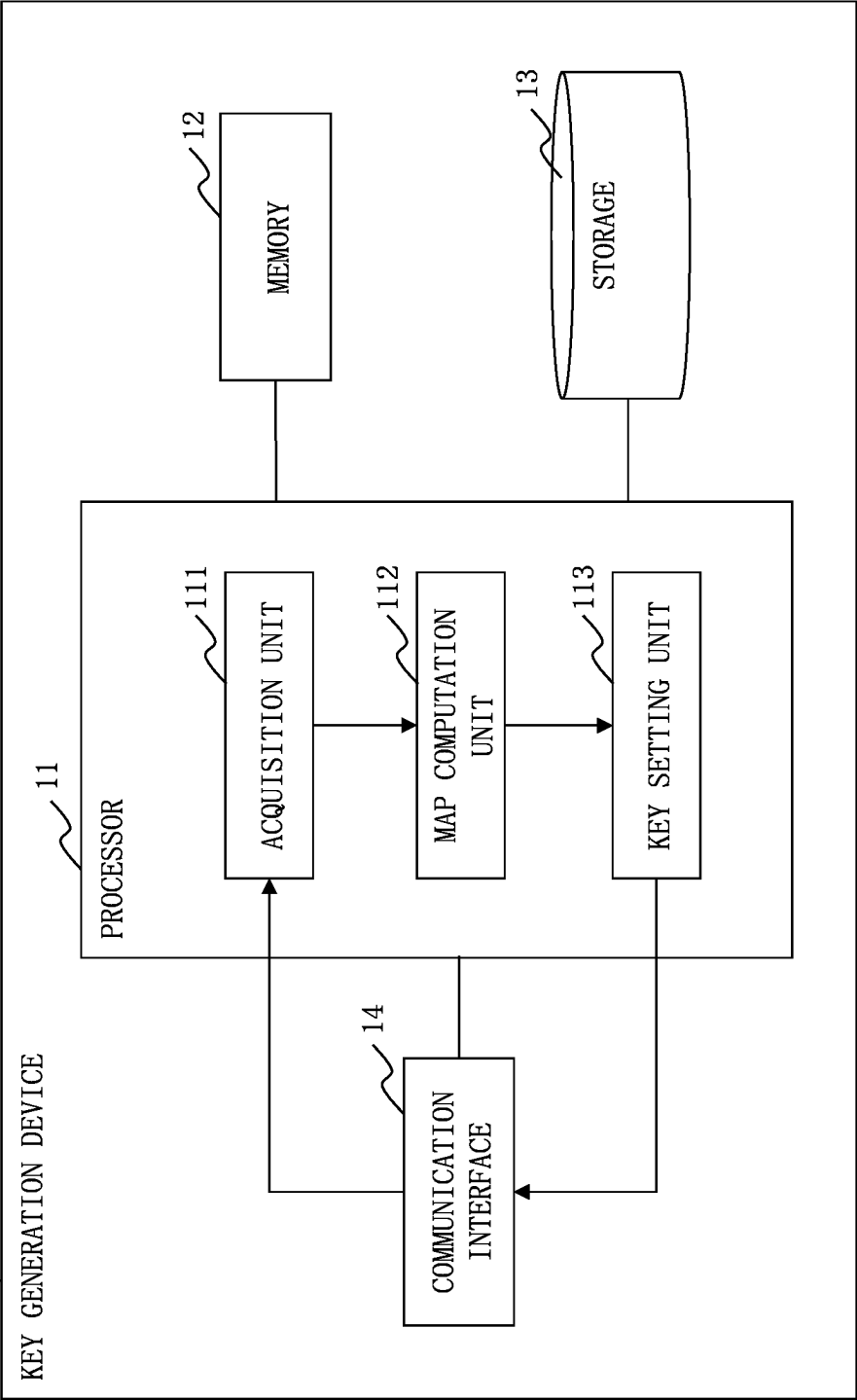


Fig. 3

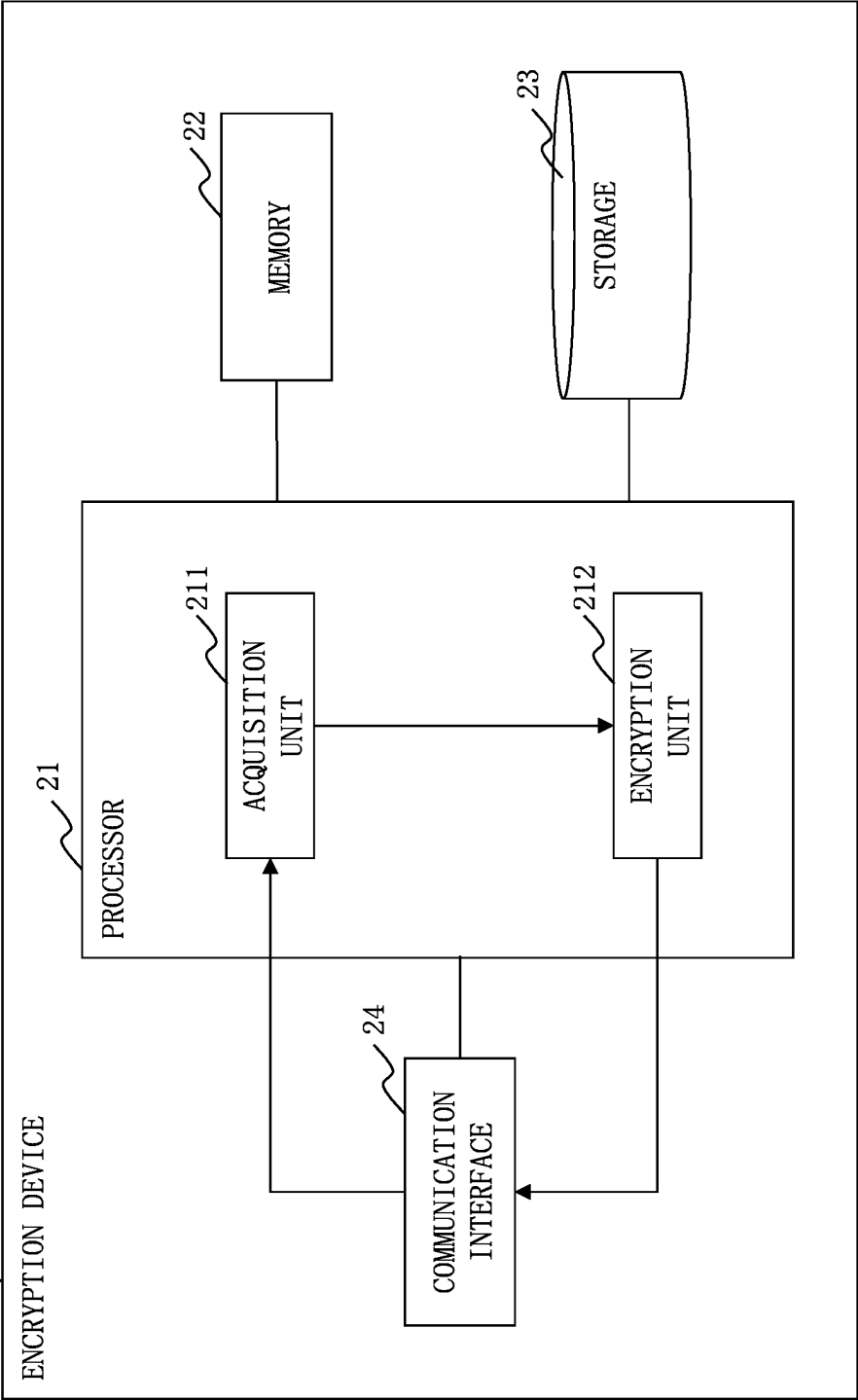
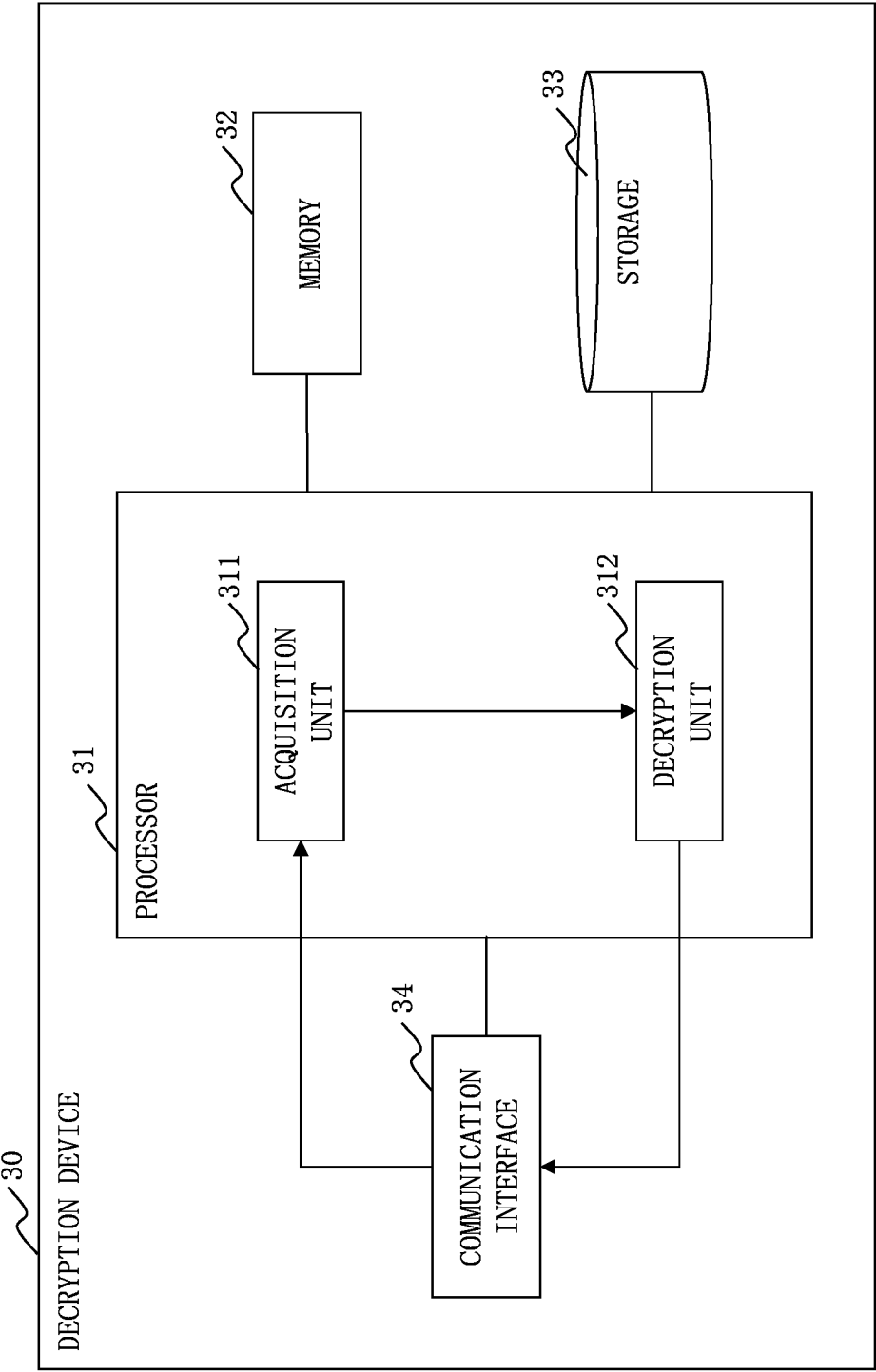


Fig. 4



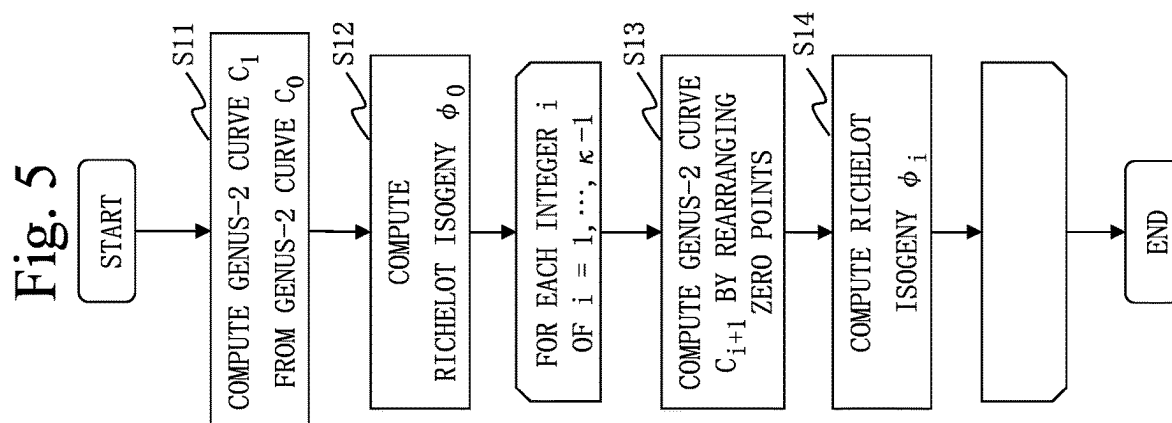


Fig. 6

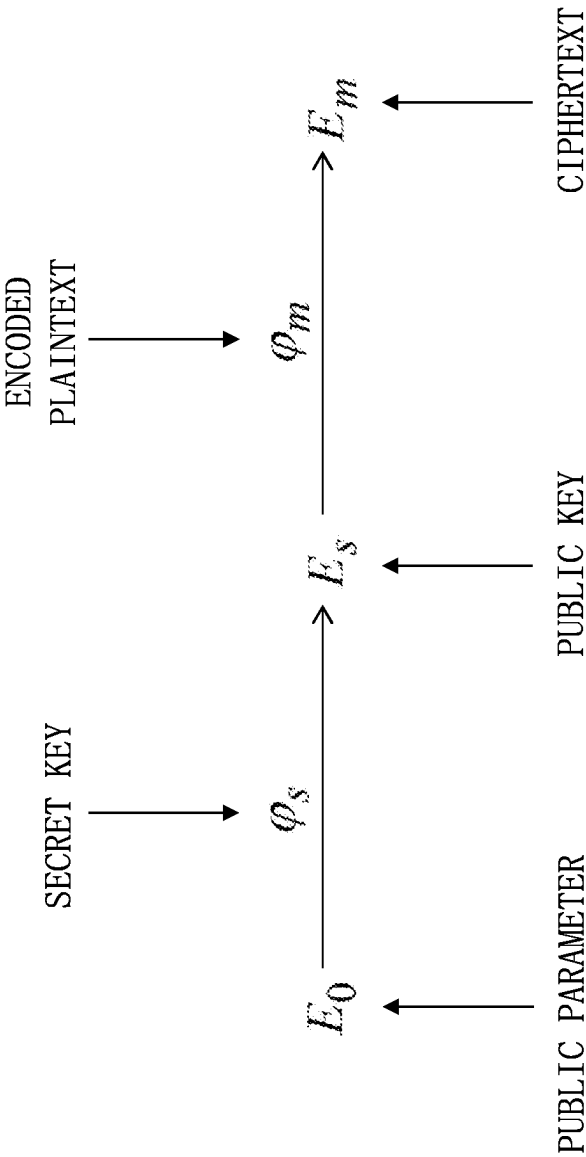


Fig. 7

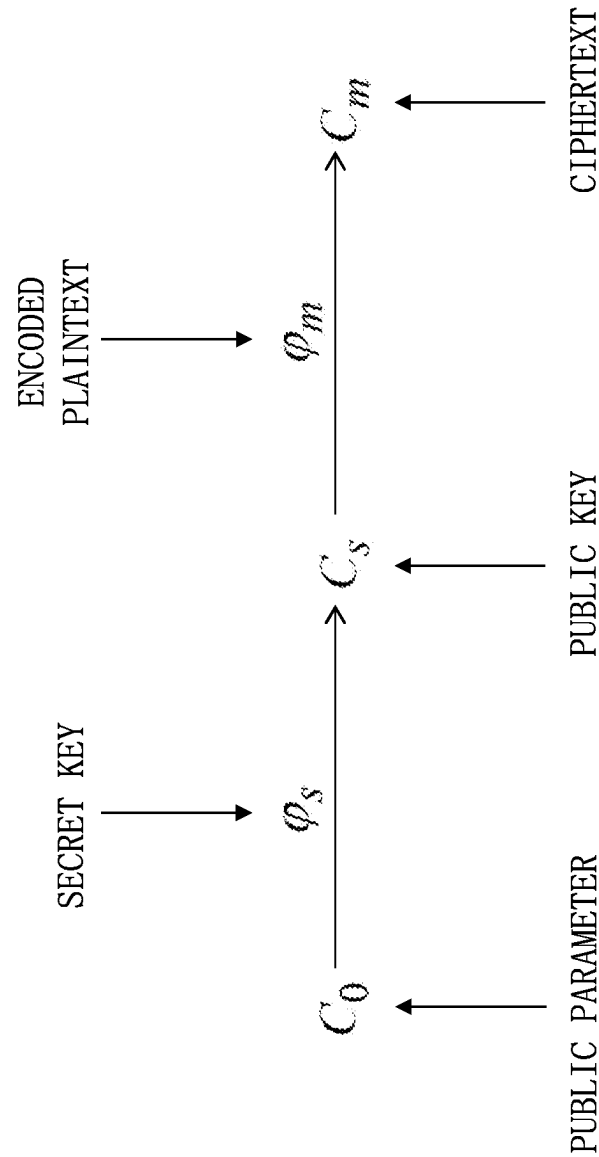


Fig. 8

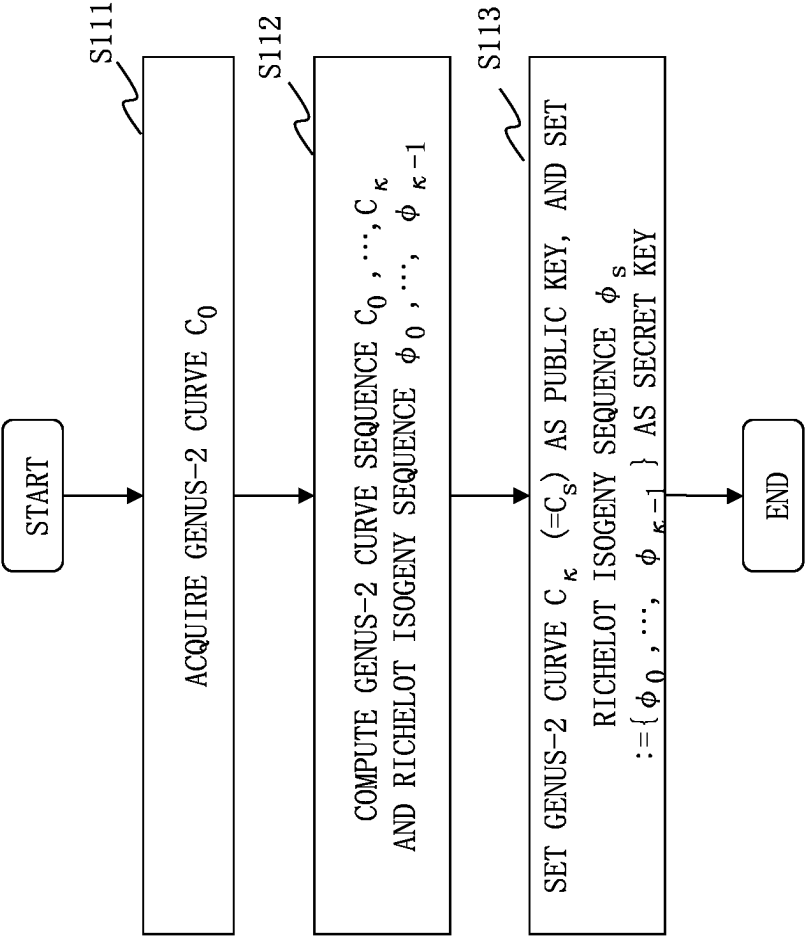


Fig. 9

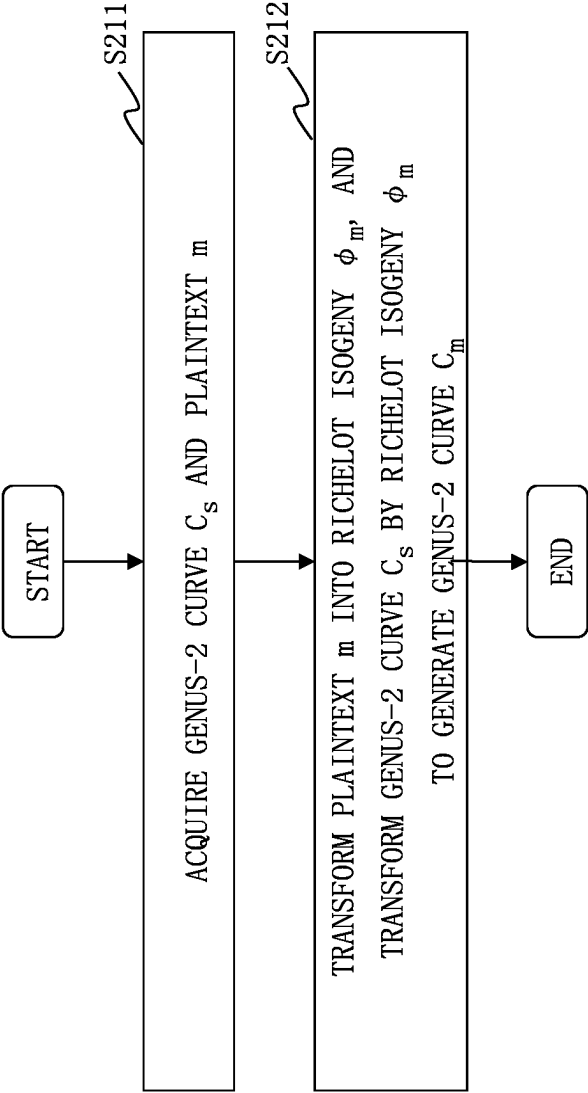


Fig. 10

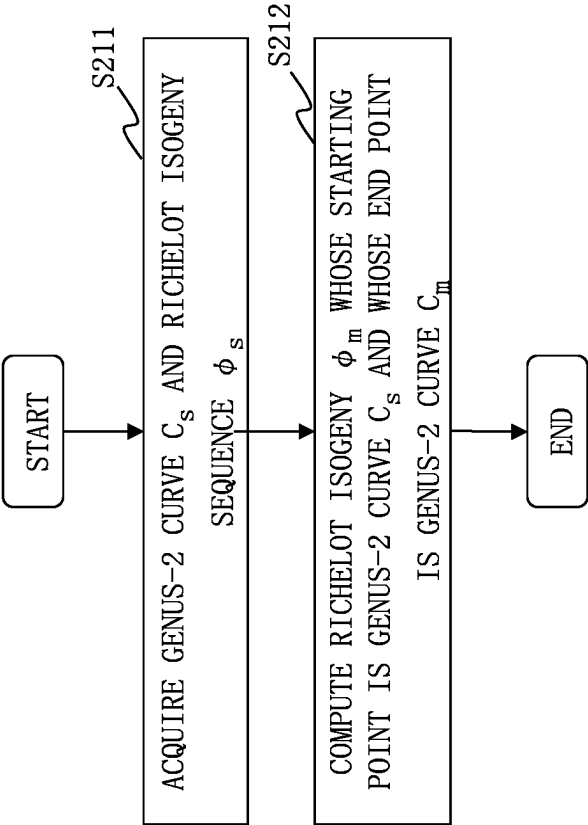


Fig. 11

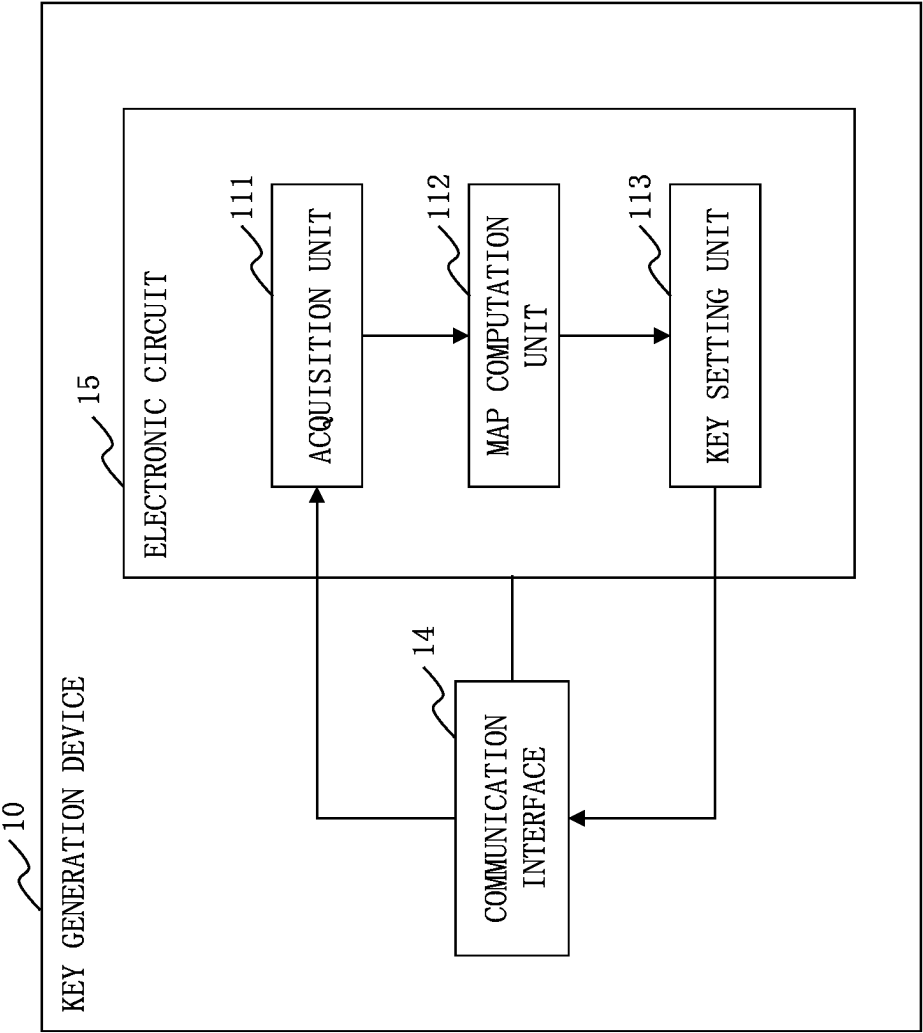


Fig. 12

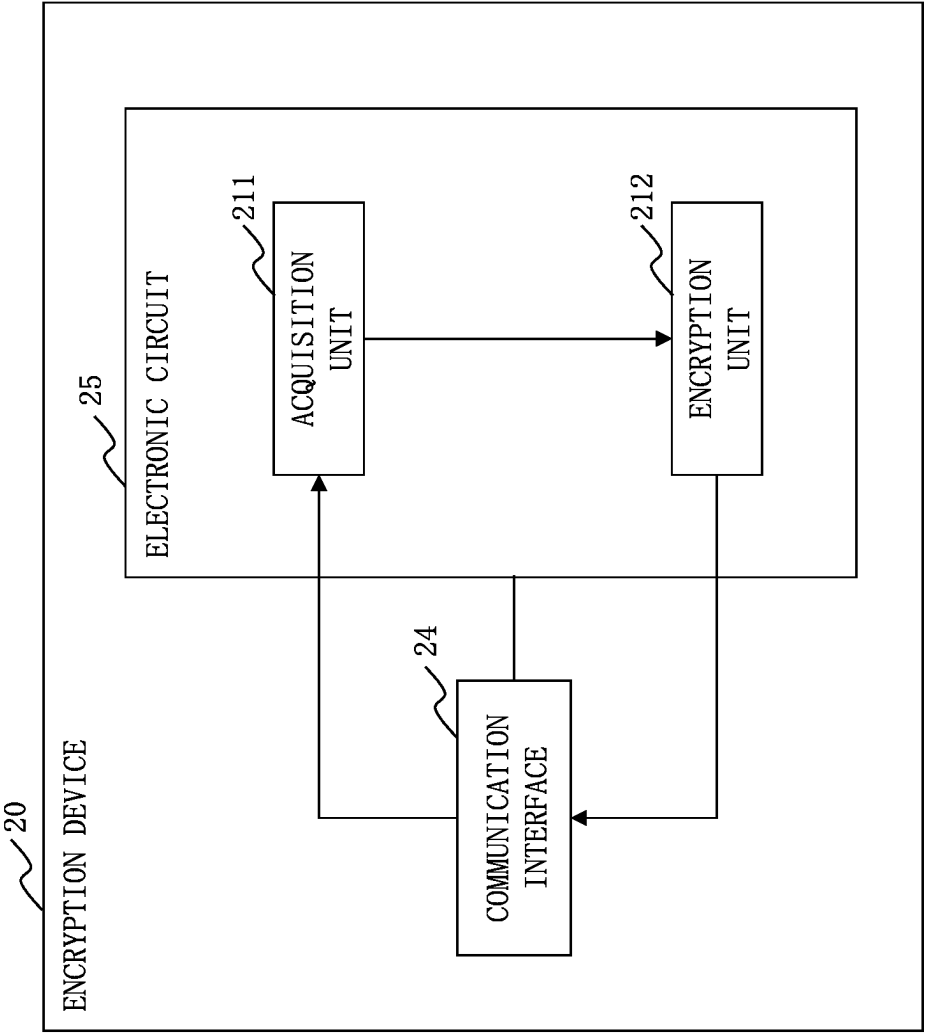
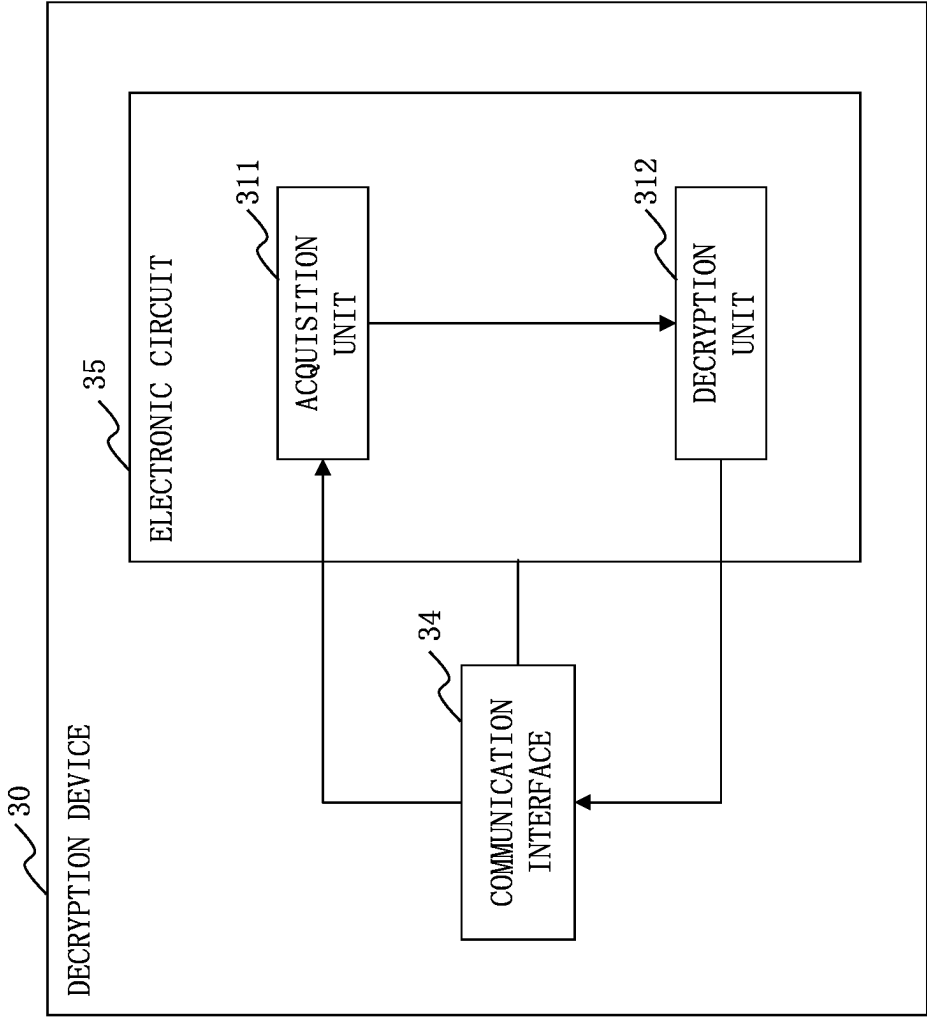


Fig. 13



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CRYPTOGRAPHIC SYSTEM, ENCRYPTION DEVICE, DECRYPTION DEVICE, AND KEY GENERATION DEVICE

CROSS REFERENCE TO RELATED APPLICATION

This application is a Continuation of PCT International Application No. PCT/JP2020/035870, filed on Sep. 23, 2020, which is hereby expressly incorporated by reference into the present application.

TECHNICAL FIELD

The present disclosure relates to isogeny-based cryptography.

BACKGROUND ART

Isogeny-based cryptography includes types such as supersingular isogeny Diffie-Hellman key exchange (SIDH) and supersingular isogeny key encapsulation (SIKE). In isogeny-based cryptography such as SIDH and SIKE, security cannot be based on the hardness of the basic isogeny problem, and security needs to be based on the hardness of the isogeny problem of the SIDH type, which requires giving an attacker (decrypting party) even additional information that is given as points of an elliptic curve. Therefore, in isogeny-based cryptography of the SIDH type, construction of a scheme whose security is based on the hardness of the basic isogeny problem without additional information has been awaited.

Non-Patent Literature 1 describes a SETA encryption scheme, which is isogeny-based cryptography of the SIDH type whose security is based on the hardness of the basic isogeny problem without additional information.

CITATION LIST

Non-Patent Literature

Non-Patent Literature 1: C. D. S. Guilhem, P. Kutas, C. Petit, J. Silva, SETA: Supersingular Encryption from Torsion Attacks.

Non-Patent Literature 2: C. Petit, Faster Algorithms for Isogeny Problems using Torsion Point Images.

SUMMARY OF INVENTION

Technical Problem

In the SETA encryption scheme, the cryptanalysis method described in Non-Patent Literature 2 is used for a decryption algorithm. Therefore, decryption requires a great amount of time.

An object of the present disclosure is to reduce the time required for decryption in the SETA encryption scheme.

Solution to Problem

A cryptographic system according to the present disclosure performs a cryptographic process in which a Richelot isogeny sequence φ_s whose starting point is an abelian surface A_0 and whose end point is an abelian surface A_s is a secret key and the abelian surface A_s is a public key, and the cryptographic system includes

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an encryption device to compute an abelian surface A_m by transitioning the abelian surface A_s , which is the public key, by a Richelot isogeny sequence φ_m generated by encoding a plaintext m , and set the abelian surface A_m as a ciphertext; and a decryption device to compute a Richelot isogeny φ_m whose starting point is the abelian surface A_s , which is the public key, and whose end point is the abelian surface A_m , which is the ciphertext, based on the Richelot isogeny sequence φ_s , which is the secret key.

Advantageous Effects of Invention

In the present disclosure, a Richelot isogeny sequence φ_s whose starting point is an abelian surface A_0 and whose end point is an abelian surface A_s is used as a secret key, and the abelian surface A_s is used as a public key. This allows the length of a prime number p in the SETA encryption scheme to be reduced to $1/3$. As a result, the time required for decryption in the SETA encryption scheme can be reduced.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a configuration diagram of a cryptographic system 1 according to a first embodiment;

FIG. 2 is a configuration diagram of a key generation device 10 according to the first embodiment;

FIG. 3 is a configuration diagram of an encryption device 20 according to the first embodiment;

FIG. 4 is a configuration diagram of a decryption device 30 according to the first embodiment;

FIG. 5 is a flowchart of a process of computing a genus-2 curve sequence $C_0, \dots, C_K$ according to the first embodiment;

FIG. 6 is a diagram explaining a SETA encryption scheme;

FIG. 7 is a diagram explaining a genus-2 SETA cryptographic scheme according to the first embodiment;

FIG. 8 is a flowchart illustrating operation of the key generation device 10 according to the first embodiment;

FIG. 9 is a flowchart illustrating operation of the encryption device 20 according to the first embodiment;

FIG. 10 is a flowchart illustrating operation of the decryption device 30 according to the first embodiment;

FIG. 11 is a configuration diagram of the key generation device 10 according to a first variation;

FIG. 12 is a configuration diagram of the encryption device 20 according to the first variation; and

FIG. 13 is a configuration diagram of the decryption device 30 according to the first variation.

DESCRIPTION OF EMBODIMENTS

First Embodiment

Description of Configurations

Referring to FIG. 1, a configuration of a cryptographic system 1 according to a first embodiment will be described.

The cryptographic system 1 includes a key generation device 10, an encryption device 20, and a decryption device 30. The key generation device 10, the encryption device 20, and the decryption device 30 are connected via a communication channel 40 such as a local area network (LAN) or the Internet.

Referring to FIG. 2, a configuration of the key generation device 10 according to the first embodiment will be described.

The key generation device **10** is a computer.

The key generation device **10** includes hardware of a processor **11**, a memory **12**, a storage **13**, and a communication interface **14**. The processor **11** is connected with other hardware components via signal lines and controls these other hardware components.

The key generation device **10** includes, as functional components, an acquisition unit **111**, a map computation unit **112**, and a key setting unit **113**. The functions of the functional components of the key generation device **10** are realized by software.

The storage **13** stores programs that realize the functions of the functional components of the key generation device **10**. These programs are read into the memory **12** by the processor **11** and executed by the processor **11**. This realizes the functions of the functional components of the key generation device **10**.

Referring to FIG. 3, a configuration of the encryption device **20** according to the first embodiment will be described.

The encryption device **20** is a computer.

The encryption device **20** includes hardware of a processor **21**, a memory **22**, a storage **23**, and a communication interface **24**. The processor **21** is connected with other hardware components via signal lines and controls these other hardware components.

The encryption device **20** includes, as functional components, an acquisition unit **211** and an encryption unit **212**. The functions of the functional components of the encryption device **20** are realized by software.

The storage **23** stores programs that realize the functions of the functional components of the encryption device **20**. These programs are read into the memory **22** by the processor **21** and executed by the processor **21**. This realizes the functions of the functional components of the encryption device **20**.

Referring to FIG. 4, a configuration of the decryption device **30** according to the first embodiment will be described.

The decryption device **30** is a computer.

The decryption device **30** includes hardware of a processor **31**, a memory **32**, a storage **33**, and a communication interface **34**. The processor **31** is connected with other hardware components via signal lines and controls these other hardware components.

The decryption device **30** includes, as functional components, an acquisition unit **311** and a decryption unit **312**. The functions of the functional components of the decryption device **30** are realized by software.

The storage **33** stores programs that realize the functions of the functional components of the decryption device **30**. These programs are read into the memory **32** by the processor **31** and executed by the processor **31**. This realizes the functions of the functional components of the decryption device **30**.

Each of the processors **11**, **21**, and **31** is an integrated circuit (IC) that performs processing. Specific examples of each of the processors **11**, **21**, and **31** are a central processing unit (CPU), a digital signal processor (DSP), and a graphics processing unit (GPU).

Each of the memories **12**, **22**, and **32** is a storage device to temporarily store data. Specific examples of each of the memories **12**, **22**, and **32** are a static random access memory (SRAM) and a dynamic random access memory (DRAM).

Each of the storages **13**, **23**, and **33** is a storage device to store data. A specific example of each of the storages **13**, **23**, and **33** is a hard disk drive (HDD). Each of the storages **13**,

23, and **33** may be a portable recording medium such as a Secure Digital (SD, registered trademark) memory card, CompactFlash (CF, registered trademark), a NAND flash, a flexible disk, an optical disc, a compact disc, a Blu-ray (registered trademark) disc, or a digital versatile disc (DVD).

Each of the communication interfaces **14**, **24**, and **34** is an interface for communicating with external devices. Specific examples of each of the communication interfaces **14**, **24**, and **34** are an Ethernet (registered trademark) port, a Universal Serial Bus (USB) port, and a High-Definition Multimedia Interface (HDMI, registered trademark) port.

FIG. 2 illustrates only one processor **11**. However, there may be a plurality of processors **11**, and the plurality of processors **11** may cooperatively execute the programs that realize the functions. Similarly, there may be a plurality of processors **21** and a plurality of processors **31**.

Description of Operation

Referring to FIGS. 5 to 10, operation of the cryptographic system **1** according to the first embodiment will be described.

A procedure for the operation of the cryptographic system **1** according to the first embodiment is equivalent to a cryptographic process according to the first embodiment. A program that realizes the operation the cryptographic system **1** according to the first embodiment is equivalent to a cryptographic program according to the first embodiment.

Description of Concept

A concept that is the basis of a cryptographic scheme realized by the cryptographic system **1** according to the first embodiment will be described.

In Non-Patent Literature 1, elliptic curves are used. The cryptographic system **1** uses genus-2 curves in place of elliptic curves.

A genus-2 curve used by the cryptographic system **1** is an algebraic curve indicated in Formula 11, where a degree $\deg(f(X))$ is 5 or 6.

$$C: Y^2 = f(X) \quad [\text{Formula 11}]$$

As indicated in Formula 12, it is assumed that genus-2 curves are transitioned with a genus-2 curve C_0 as the starting point and a genus-2 curve C_κ as the end point by repeating Richelot isogenies κ times, where κ is an integer of 2 or more.

$$C_0 \xrightarrow{\varphi_0} C_1 \xrightarrow{\varphi_1} \dots \xrightarrow{\varphi_{\kappa-2}} C_{\kappa-1} \xrightarrow{\varphi_{\kappa-1}} C_\kappa \quad [\text{Formula 12}]$$

As the basic problem concerning cryptographic security, the cryptographic system **1** uses finding (computing) a Richelot isogeny sequence $\varphi_0, \varphi_1, \dots, \varphi_{\kappa-1}$ that interpolates the genus-2 curves C_0 and C_κ when two genus-2 curves of the genus-2 curves C_0 and C_κ are given. The cryptographic system **1** constructs genus-2 SETA cryptography that is based on the hardness of this basic problem.

Each genus-2 curve is expressed using three polynomials $G_j(X)$ with $\deg(G_j(X)) \leq 2$, as indicated in Formula 13. Note that $\deg(G_j(X))$ is the degree of a polynomial $G_j(X)$.

$$C: Y^2 = f(X) = \prod_{j=0}^2 G_j(X) \quad [\text{Formula 13}]$$

Three new polynomials $G_j^-(X)$ are defined as indicated in Formula 14.

$$\begin{aligned} [G_{j+1}(X), G_{j+2}(X)] &:= G_{j+1}'(X)G_{j+2}(X) - G_{j+2}'(X)G_{j+1}(X), \\ G_j^-(X) &:= \tau_j^{-1}[G_{j+1}(X), G_{j+2}(X)] \end{aligned} \quad [\text{Formula 14}]$$

Note that τ_j is the highest-degree coefficient of $[G_{j+1}(X), G_{j+2}(X)]$. Therefore, $G_j^-(X)$ is a polynomial whose highest-

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degree coefficient is 1. $G'_{j+1}(X)$ is a derivative of $G_{j+1}(X)$ and $G'_{j+2}(X)$ is a derivative of $G_{j+2}(X)$. Indices are assigned on the assumption that the three polynomials G_j are treated as a permutation such that $G_{j+1}(X)$ when $j=0$ is $G_1(X)$, $G_{j+1}(X)$ when $j=1$ is $G_2(X)$, $G_{j+1}(X)$ when $j=2$ is $G_0(X)$. Similarly, $G_{j+2}(X)$ when $j=0$ is $G_2(X)$, $G_{j+2}(X)$ when $j=1$ is $G_0(X)$, and $G_{j+2}(X)$ when $j=2$ is $G_1(X)$.

A new genus-2 curve C^- is defined as indicated in Formula 15.

$$\tilde{C}: Y^2 = \tilde{f}(X) = d \prod_{j=0}^2 G_j(X) \quad [\text{Formula 15}]$$

with $\tilde{d} = d \cdot \tau_0 \tau_1 \tau_2 \cdot \det((g_{j,k})_{0 \leq j,k \leq 2})^{-1}$

A correspondence (mapping) from the genus-2 curve C to the genus-2 curve C^- is called a Richelot isogeny. The Richelot isogeny from the genus-2 curve C is determined based on a division $(G_0(X), G_1(X), G_2(X))$ of $f(X)$. This corresponds to a division of zero points of $f(X)$ into three pairs (a_0, a_1) , (a_2, a_3) , and (a_4, a_5) . Note that g will be described later.

****Description of a Method of Computing a Genus-2 Curve Sequence****

The cryptographic system **1** computes a genus-2 curve sequence $C_0, \dots, C_\kappa$ by repeating Richelot isogenies from a genus-2 curve C_0 . However, computing a Richelot isogeny based on three polynomials $G_j(X)$ will revert to the genus-2 curve C . Therefore, the cryptographic system **1** computes a Richelot isogeny once, then rearranges each root (zero point) of each polynomial $G_j(X)$ for a change into three new polynomials, and then computes a Richelot isogeny.

Referring to FIG. 5, a process of computing the genus-2 curve sequence $C_0, \dots, C_\kappa$ according to the first embodiment will be described.

A genus-2 curve C_i is defined as indicated in Formula 16.

$$C_i / \mathbb{F}_p: Y^2 = f_i(X) = d \prod_{j=0}^2 G_{i,j}(X) \text{ where} \quad [\text{Formula 16}]$$

$$G_{i,j} = \sum_{k=0}^2 g_{i,j,k} X^k$$

$$= \begin{cases} (X - a_{i,2j})(X - a_{i,2j+1}) & \text{if } \deg(G_{i,j}) = 2 \\ X - a_{i,2j} \text{ or } X - a_{i,2j+1} & \text{if } \deg(G_{i,j}) = 1 \end{cases}$$

In step **S11**, three new polynomials $(G_{i,j}(X))_{j=0,1,2}$ are generated from the genus-2 curve C_0 in accordance with Formula 14. By this, a genus-2 curve C_1 is computed.

In step **S12**, as indicated in Formula 17, three polynomials $G_{i+1,j}$ are computed to compute a Richelot isogeny ϕ_0 . Note that $\tau_{0,j}$ is the highest-degree coefficient of $[G_{0j+1}(X), G_{0j+2}(X)]$.

$$\tilde{G}_{i+1,j}(X) := \tau_{0,j}^{-1} [G_{0j+1}(X), G_{0j+2}(X)] \text{ for } j=0,1,2 \quad [\text{Formula 17}]$$

Then, processes of step **S13** and step **S14** are performed for each integer i of $i=1, \dots, \kappa-1$ in ascending order.

In step **S13**, zero points $(a_{i,m})_{m=0, \dots, 5}$ of $f_i(X)$ are rearranged. Then, three new polynomials $(G_{i,j}(X))_{j=0,1,2}$ are generated in accordance with Formula 14. By this, a genus-2 curve C_{i+1} is computed.

In step **S14**, as indicated in Formula 18, three polynomials $G_{i+1,j}$ are computed to compute a Richelot isogeny ϕ_i . Note that $\tau_{i,j}$ is the highest-degree coefficient of $[G_{i,j+1}(X), G_{i,j+2}(X)]$.

$$\tilde{G}_{i+1,j}(X) := \tau_{i,j}^{-1} [G_{i,j+1}(X), G_{i,j+2}(X)] \text{ for } j=0,1,2 \quad [\text{Formula 18}]$$

****SETA Encryption Scheme****

Referring to FIG. 6, a SETA encryption scheme described in Non-Patent Literature 1 will be described.

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In the SETA encryption scheme, elliptic curves E_0, E_s , and E_m are used. The elliptic curve E_0 is a public parameter. The elliptic curve E_0 is a special curve whose endomorphism ring is easy.

<Key Generation Process>

First, an isogeny sequence ϕ_s , which is a secret key, is generated. Then, the elliptic curve E_0 is transitioned by the isogeny sequence ϕ_s to calculate an elliptic curve E_s , and the elliptic curve E_s is set as a public key.

<Encryption Process>

First, a plaintext m is suitably encoded and transformed into an isogeny sequence ϕ_m . Then, the elliptic curve E_s , which is the public key, is transitioned by the isogeny sequence ϕ_m to compute an elliptic curve E_m . The elliptic curve E_m is set as a ciphertext.

<Decryption Process>

An endomorphism ring of the elliptic curve E_s can be computed through computing an endomorphism ring of the elliptic curve E_0 , using the isogeny sequence ϕ_s , which is the secret key. Therefore, an isogeny problem in which the elliptic curve E_s is the starting point and the elliptic curve E_m is the end point can be solved in polynomial time, and the isogeny sequence ϕ_m can be obtained. The plaintext m is computed from the isogeny sequence ϕ_m by performing a decode operation, which is the inverse operation of an encode operation at the time of encryption.

The point of decryption is that a user who knows the isogeny sequence ϕ_s , which is the secret key, can reduce computing the endomorphism ring of the elliptic curve E_s to computing the endomorphism ring of the elliptic curve E_0 through the isogeny sequence ϕ_s .

****Genus-2 SETA Cryptographic Scheme****

Referring to FIGS. 7 to 10, a genus-2 SETA cryptographic scheme that is realized by the cryptographic system **1** according to the first embodiment will be described.

As indicated in FIG. 7, the genus-2 SETA cryptographic scheme is realized by replacing elliptic curves E_0, E_s , and E_m in the SETA encryption scheme with genus-2 curves C_0, C_s , and C_m , respectively, and replacing isogeny sequences ϕ_s and ϕ_m with Richelot isogeny sequences ϕ_s and ϕ_m , respectively.

Referring to FIG. 8, operation of the key generation device **10** according to the first embodiment will be described.

A procedure for the operation of the key generation device **10** according to the first embodiment is equivalent to a key generation method according to the first embodiment. A program that realizes the operation of the key generation device **10** according to the first embodiment is equivalent to a key generation program according to the first embodiment. (Step **S111**: Acquisition Process)

The acquisition unit **111** acquires a genus-2 curve C_0 , which is a public parameter.

The acquisition unit **111** writes the genus-2 curve C_0 in the memory **12**.

(Step **S112**: Map Computation Process)

The map computation unit **112** computes a Richelot isogeny ϕ_m of the genus-2 curve C_0 acquired in step **S111** to compute a genus-2 curve C_1 . The map computation unit **112** computes a Richelot isogeny ϕ_i to compute a genus-2 curve C_{i+1} by rearranging zero points of a genus-2 curve C_i for each integer i of $i=1, \dots, \kappa-1$ in ascending order, using an integer κ of 2 or more.

Specifically, the map computation unit **112** computes a genus-2 curve sequence $C_0, \dots, C_\kappa$ and a Richelot isogeny sequence $\phi_0, \dots, \phi_{\kappa-1}$ by performing the process of

computing of the genus-2 curve sequence $C_0, \dots, C_K$ described with reference to FIG. 5.

The map computation unit 112 writes the genus-2 curve sequence $C_0, \dots, C_K$ and the Richelot isogeny sequence $\varphi_0, \dots, \varphi_{K-1}$ in the memory 12.
(Step S113: Key Setting Process)

The key setting unit 113 sets the genus-2 curve C_K computed in step S112 as a genus-2 curve C_s as a public key. The key setting unit 113 sets the Richelot isogeny sequence $\varphi_s := \{\varphi_0, \dots, \varphi_{K-1}\}$ computed in step S112 as a secret key. That is, the Richelot isogeny sequence φ_s whose starting point is the genus-2 curve C_0 and whose end point is the genus-2 curve C_s is set as the secret key.

The key setting unit 113 transmits the public key to the encryption device 20 and the decryption device 30 via the communication interface 14. The key setting unit 113 transmits the secret key in secrecy to the decryption device 30 via the communication interface 14. To transmit in secrecy means, for example, to transmit after encryption using an existing encryption scheme.

The secret key is the Richelot isogeny sequence φ_s . Substantially, this means that a method of permutations $\sigma_0, \dots, \sigma_{K-1}$, which is how roots are rearranged in the process of step 13 in the process of computing the genus-2 curve sequence $C_0, \dots, C_K$, corresponds to the secret key.

Referring to FIG. 9, operation of the encryption device 20 according to the first embodiment will be described.

A procedure for the operation of the encryption device 20 according to the first embodiment is equivalent to an encryption method according to the first embodiment. A program that realizes the operation of the encryption device 20 according to the first embodiment is equivalent to an encryption program according to the first embodiment.
(Step S211: Acquisition Process)

The acquisition unit 211 acquires the genus-2 curve C_s , which is the public key generated by the key generation device 10. The acquisition unit 211 acquires a plaintext m . The plaintext m is input by a user of the encryption device 20.

The acquisition unit 211 writes the genus-2 curve C_s and the plaintext m in the memory 22.

(Step S212: Encryption Process)

The encryption unit 212 encodes the plaintext m acquired in step S211 to transform the plaintext m into a Richelot isogeny sequence φ_m . The encryption unit 212 transitions the genus-2 curve C_s , which is the public key, acquired in step S211 by the Richelot isogeny sequence φ_m to compute a genus-2 curve C_m . Then, the encryption unit 212 sets the genus-2 curve C_m as a ciphertext.

The encryption unit 212 transmits the ciphertext to the decryption device 30 via the communication interface 24.

Referring to FIG. 10, operation of the decryption device 30 according to the first embodiment will be described.

A procedure for the operation of the decryption device 30 according to the first embodiment is equivalent to a decryption method according to the first embodiment. A program that realizes the operation of the decryption device 30 according to the first embodiment is equivalent to a decryption program according to the first embodiment.
(Step S311: Acquisition Process)

The acquisition unit 311 acquires the public parameter, and acquires the public key and the secret key generated by the key generation device 10. The acquisition unit 311 acquires the ciphertext generated by the encryption device 20.

The acquisition unit 311 writes the public parameter, the public key, the secret key, and the ciphertext in the memory 32.

(Step S312: Decryption Process)

The decryption unit 312 computes a Richelot isogeny φ_m whose starting point is the genus-2 curve C_s , which is the public key, acquired in step S311, and whose end point is the genus-2 curve C_m , which is the ciphertext, acquired in step S311, based on the Richelot isogeny sequence φ_s , which is the secret key, and the genus-2 curve C_0 , which is the public key, acquired in step S311. Then, the decryption unit 312 decodes the Richelot isogeny φ_m to compute a plaintext $m' = m$. This decoding is the inverse operation of the encoding performed in step S212.

The decryption unit 312 outputs the plaintext m' via the communication interface 14.

Effects of the First Embodiment

As described above, the cryptographic system 1 according to the first embodiment realizes the genus-2 SETA cryptographic scheme by replacing the elliptic curves E_0, E_s , and E_m in the SETA encryption scheme with the genus-2 curves C_0, C_s , and C_m , and replacing the isogeny sequences φ_s and φ_m with the Richelot isogeny sequences φ_s and φ_m .

In the genus-2 SETA cryptographic scheme, the value of the prime number p used in the SETA encryption scheme can be reduced to $1/3$. As a result, the time required for decryption can be reduced.

Other Configurations

<First Variation>

In the first embodiment, the case where the genus-2 curves C_0, C_s , and C_m are used has been described. However, the genus-2 curves C_0, C_s , and C_m used by the cryptographic system 1 according to the first embodiment can be interpreted as abelian surfaces A_0, A_s , and A_m . That is, a cryptographic scheme having substantially the same effects as those of the genus-2 SETA cryptographic scheme can be realized by replacing the elliptic curves E_0, E_s , and E_m in the SETA encryption scheme with the abelian surfaces A_0, A_s , and A_m , respectively, and replacing the isogeny sequences φ_s and φ_m with the Richelot isogeny sequences φ_s and φ_m , respectively.

When a genus-2 curve is used as in the first embodiment, it is desirable that the genus-2 curve be the genus-2 curve corresponding to a portion adjacent to a portion decomposed into the direct product of the elliptic curve E_0 out of portions not decomposed into the direct product of the elliptic curve E_0 in the abelian surface.

When a genus-2 curve is not used, it is desirable that a portion decomposed into the direct product of the elliptic curve be used.

The elliptic curve E_0 used in the direct product of the elliptic curve that decomposes the abelian surface is $E_0/F_p: y^2 = x^3 + x$ when the remainder of dividing the prime number p by 4 is 3, or $E_0/F_p: y^2 = x^3 + c$ when the remainder of dividing the prime number p by 4 is 1, where c is a constant. Note that F_p is a field modulo the prime number p .

<Second Variation>

In the first embodiment, the functional components are realized by software. As a second variation, the functional components may be realized by hardware. With regard to this second variation, differences from the first embodiment will be described.

Referring to FIG. 11, a configuration of the key generation device 10 according to the second variation will be described.

When the functional components are realized by hardware, the key generation device **10** includes an electronic circuit **15** in place of the processor **11**, the memory **12**, and the storage **13**. The electronic circuit **15** is a dedicated circuit that realizes the functions of the functional components, the memory **12**, and the storage **13**.

Referring to FIG. **12**, a configuration of the encryption device **20** according to the second variation will be described.

When the functional components are realized by hardware, the encryption device **20** includes an electronic circuit **25** in place of the processor **21**, the memory **22**, and the storage **23**. The electronic circuit **25** is a dedicated circuit that realizes the functions of the functional components, the memory **22**, and the storage **23**.

Referring to FIG. **13**, a configuration of the decryption device **30** according to the second variation will be described.

When the functional components are realized by hardware, the decryption device **30** includes an electronic circuit **35** in place of the processor **31**, the memory **32**, and the storage **33**. The electronic circuit **35** is a dedicated circuit that realizes the functions of the functional components, the memory **32**, and the storage **33**.

Each of the electronic circuits **15**, **25**, and **35** is assumed to be a single circuit, a composite circuit, a programmed processor, a parallel-programmed processor, a logic IC, a gate array (GA), an application specific integrated circuit (ASIC), or a field-programmable gate array (FPGA).

The functional components may be realized by one electronic circuit **15**, one electronic circuit **25**, and one electronic circuit **35**, or the functional components may be distributed among and realized by a plurality of electronic circuits **15**, a plurality of electronic circuits **25**, and a plurality of electronic circuits **35**.

<Third Variation>

As a third variation, some of the functional components may be realized by hardware, and the rest of the functional components may be realized by software.

Each of the processors **11**, **21**, **31**, the memories **12**, **22**, **32**, the storages **13**, **23**, **33**, the electronic circuits **15**, **25**, and **35** is referred to as processing circuitry. That is, the functions of the functional components are realized by the processing circuitry.

The embodiment and variations of the present disclosure have been described above. Two or more of these embodiment and variations may be implemented in combination. Alternatively, one or more of them may be partially implemented. The present disclosure is not limited to the above embodiment and variations, and various modifications can be made as needed.

REFERENCE SIGNS LIST

1: cryptographic system, **10**: key generation device, **11**: processor, **12**: memory, **13**: storage, **14**: communication interface, **15**: electronic circuit, **111**: acquisition unit, **112**: map computation unit, **113**: key setting unit, **20**: encryption device, **21**: processor, **22**: memory, **23**: storage, **24**: communication interface, **25**: electronic circuit, **211**: acquisition unit, **212**: encryption unit, **30**: decryption device, **31**: processor, **32**: memory, **33**: storage, **34**: communication interface, **35**: electronic circuit, **311**: acquisition unit, **312**: decryption unit.

The invention claimed is:

1. A cryptographic system to perform a cryptographic process in which a Richelot isogeny sequence φ_s includes a

first starting point and a first end point, the first starting point being an abelian surface A_0 and the first end point being an abelian surface A_s , the abelian surface A_0 being a first secret key and the abelian surface A_s being a first public key, the cryptographic system comprising:

processing circuitry configured to:

compute an abelian surface A_m by transitioning the abelian surface A_s , which is the first public key, by a Richelot isogeny sequence φ_m generated by encoding a plaintext m , and set the abelian surface A_m as a ciphertext; and

compute a Richelot isogeny φ_m with a second start point and a second end point, the second starting point being the abelian surface A_s , the abelian surface A_s being a second public key, and the second end point is the abelian surface A_m , the abelian surface A_m being the ciphertext, based on the Richelot isogeny sequence φ_s , the Richelot isogeny sequence φ_s bring a second secret key.

2. The cryptographic system according to claim **1**, wherein the processing circuitry is further configured to:

compute a Richelot isogeny φ_0 of an abelian surface A_0 to compute an abelian surface A_1 , compute a Richelot isogeny φ_i to compute an abelian surface A_{i+1} by rearranging zero points of the abelian surface A_i for each integer i of $i=1, \dots, \kappa-1$ in ascending order, using an integer κ of 2 or more, set the abelian surface A_s that is an abelian surface A_κ as a third public key, and set the Richelot isogeny sequence φ_s that is a set of Richelot isogenies φ_i for integers i of $i=0, \dots, \kappa-1$ as a third secret key.

3. The cryptographic system according to claim **1**, wherein the processing circuitry is configured to use, as the abelian surface A_0 , a genus-2 curve corresponding to the abelian surface A_0 .

4. The cryptographic system according to claim **2**, wherein the processing circuitry is configured to use, as the abelian surface A_0 , a genus-2 curve corresponding to the abelian surface A_0 .

5. The cryptographic system according to claim **3**, wherein the genus-2 curve corresponds to a portion adjacent to a portion decomposed into a direct product of an elliptic curve E_0 out of portions not decomposed into the direct product of the elliptic curve E_0 in the abelian surface A_0 .

6. The cryptographic system according to claim **4**, wherein the genus-2 curve corresponds to a portion adjacent to a portion decomposed into a direct product of an elliptic curve E_0 out of portions not decomposed into the direct product of the elliptic curve E_0 in the abelian surface A_0 .

7. The cryptographic system according to claim **1**, wherein the processing circuitry is configured to use, as the abelian surface A_0 , a portion decomposed into a direct product of an elliptic curve E_0 in the abelian surface A_0 .

8. The cryptographic system according to claim **2**, wherein the processing circuitry is configured to use, as the abelian surface A_0 , a portion decomposed into a direct product of an elliptic curve E_0 in the abelian surface A_0 .

9. The cryptographic system according to claim **5**, wherein the elliptic curve E_0 is $E_0/F_p: y^2=x^3+x$ when a remainder of dividing a prime number p by 4 is 3, or $E_0/F_p: y^2=x^3+c$ when the remainder of dividing the prime number p by 4 is 1, where c is a constant, and F_p is a field modulo a prime number p .

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10. The cryptographic system according to claim 6, wherein the elliptic curve E_0 is $E_0/F_p: y^2=x^3+ax$ when a remainder of dividing a prime number p by 4 is 3, or $E_0/F_p: y^2=x^3+c$ when the remainder of dividing the prime number p by 4 is 1, where c is a constant, and F_p is a field modulo a prime number p . 5
11. The cryptographic system according to claim 7, wherein the elliptic curve E_0 is $E_0/F_p: y^2=x^3+ax$ when a remainder of dividing a prime number p by 4 is 3, or $E_0/F_p: y^2=x^3+c$ when the remainder of dividing the prime number p by 4 is 1, where c is a constant, and F_p is a field modulo a prime number p . 10
12. The cryptographic system according to claim 8, wherein the elliptic curve E_0 is $E_0/F_p: y^2=x^3+ax$ when a remainder of dividing a prime number p by 4 is 3, or $E_0/F_p: y^2=x^3+c$ when the remainder of dividing the prime number p by 4 is 1, where c is a constant, and F_p is a field modulo a prime number p . 15
13. An encryption device in a cryptographic system to perform a cryptographic process in which a Richelot isogeny sequence φ_s includes a first starting point and a first end point, the first starting point being an abelian surface A_0 and the first end point being an abelian surface A_s , the abelian surface A_0 being a first secret key and the abelian surface A_s being a first public key, the encryption device comprising processing circuitry configured to compute an abelian surface A_m by transitioning the abelian surface A_s , the abelian surface A_s being the first public key, by a Richelot isogeny sequence φ_m generated by encoding a plaintext m , and set the abelian surface A_m as a ciphertext. 20 25 30
14. A decryption device in a cryptographic system to perform a cryptographic process in which a Richelot isogeny φ_s includes a first starting point and a first end point, the first

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- starting point being an abelian surface A_0 and the first end point being an abelian surface A_s , the abelian surface A_0 being a first secret key and the abelian surface A_s being a first public key, the decryption device comprising processing circuitry configured to:
- acquire a ciphertext that is an abelian surface A_m computed by transitioning the abelian surface A_s , the abelian surface A_s being a first public key, by a Richelot isogeny sequence φ_m generated by encoding a plaintext m ; and
 - compute a Richelot isogeny φ_m with a second start point and a second end point, the second starting point being the abelian surface A_s , the abelian surface A_s being a second public key, and the second end point is the abelian surface A_m , the abelian surface A_m being the ciphertext, based on the Richelot isogeny sequence φ_s , the Richelot isogeny sequence φ_s being a second secret key.
15. A key generation device comprising processing circuitry configured to:
- acquire an abelian surface A_0 , which is a public parameter;
 - compute a Richelot isogeny φ_0 of the acquired abelian surface A_0 to compute an abelian surface A_1 , and compute a Richelot isogeny φ_i to compute an abelian surface A_{i+1} by rearranging zero points of the abelian surface A_i for each integer i of $i=1, \dots, \kappa-1$ in ascending order, using an integer κ of 2 or more; and
 - set a computed abelian surface A_κ as a public key, and set a Richelot isogeny sequence that is a set of Richelot isogenies φ_i for integers i of $i=0, \dots, \kappa-1$ as a secret key.

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